LAPORAN PRAKTIKUM
KONFIGURASI DHCP

OLEH:

AISA SILVANA S. NABABAN

43324011

D3 TEKNOLOGI KOMPUTER

INSTITUT TEKNOLOGI DEL

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1. Pastikan VM mempunyai 2 NIC/Network Adapter. Adapter pertama menggunakan bridge, ini digunakan untuk mengkoneksikan ubuntu server ke internet, sehingga ip pada Network adapter ini akan secara otomatis diberikan (Dynamic). Adapter kedua menggunakan custom (contohnya VMnet1) dan ip pada adapter ini akan di setting secara manual (Static).

Device	Summary
Memory	4 GB
Processors	2
Hard Disk (SCSI)	40 GB
CD/DVD (SATA)	Using file E:\Virtual Machines...
Network Adapter	Bridged (Automatic)
Network Adapter 2	Custom (VMnet1)
USB Controller	Present
Sound Card	Auto detect
Display	Auto detect

2. Cek ip nya, harus mendapatkan IP untuk NIC nya. Jika belum ada maka perlu di setting ip secara static. Pergi ke dirketori /etc/netplan. Lalu edit file .yaml yang ada pada direktori itu.

```
  GNU nano 7.2                                          00-installer-config.yaml *
network:
  version: 2
  ethernets:
    ens33:
      dhcp4: true
    ens37:
      dhcp4: no
      addresses:
        - 192.168.100.1/24
```

3. Simpan file lalu jalankan sudo netplan apply perintah ini akan menerapkan konfigurasi IP baru

```
aisasilvana@aisasilvana:/etc/netplan$ sudo netplan apply
aisasilvana@aisasilvana:/etc/netplan$
```

4. Verifikasi IP Address Setelah Netplan Apply cek kembali alamat IP dengan perintah berikut Tujuannya untuk memastikan konfigurasi IP static sudah berhasil diterapkan.

```
aisasilvana@aisasilvana:/etc/netplan$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:ff:f9:78 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet6 fe80::20c:29ff:feff:f978/64 scope link
       valid_lft forever preferred_lft forever
3: ens37: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:ff:f9:82 brd ff:ff:ff:ff:ff:ff
    altname enp2s5
    inet 192.168.100.1/24 brd 192.168.100.255 scope global ens37
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:feff:f982/64 scope link
       valid_lft forever preferred_lft forever
aisasilvana@aisasilvana:/etc/netplan$
```

5. Kemudian konfigurasi DHCP Server. Edit file utama DHCP (sudo nano /etc/dhcp/dhcpd.conf) lalu tambahkan konfigurasinya seperti gambar dibawah ini

```
  GNU nano 7.2                                          /etc/dhcp/dhcpd.conf *
#subnet 10.254.239.32 netmask 255.255.255.224 {
#  range dynamic-bootp 10.254.239.40 10.254.239.60;
#  option broadcast-address 10.254.239.31;
#  option routers rtr-239-32-1.example.org;
#}

# A slightly different configuration for an internal subnet.
subnet 192.168.100.0 netmask 255.255.255.0 {
  range 192.168.100.10 192.168.100.100;
  option domain-name-servers 192.168.100.1;
  option domain-name "internal.example.org";
  option subnet-mask 255.255.255.0;
  option routers 192.168.100.1;
  option broadcast-address 192.168.100.255;
  default-lease-time 600;
  max-lease-time 7200;
}
# Hosts which require special configuration options can be listed in
# host statements.   If no address is specified, the address will be
# allocated dynamically (if possible), but the host-specific information
# will still come from the host declaration.

#host passacaglia {
#  hardware ethernet 0:0:c0:5d:bd:95;
#  filename "vmunix.passacaglia";
#  server-name "toccata.example.com";
#}
```

6. Kemudian tentukan Interface DHCP edit file konfigurasi Ubah menjadi: INTERFACESv4="ens37" Karena ens37 adalah interface yang digunakan untuk melayani client.

```
  GNU nano 7.2                                          /etc/default/isc-dhcp-server *
# Defaults for isc-dhcp-server (sourced by /etc/init.d/isc-dhcp-server)

# Path to dhcpd's config file (default: /etc/dhcp/dhcpd.conf).
#DHCPDv4_CONF=/etc/dhcp/dhcpd.conf
#DHCPDv6_CONF=/etc/dhcp/dhcpd6.conf

# Path to dhcpd's PID file (default: /var/run/dhcpd.pid).
#DHCPDv4_PID=/var/run/dhcpd.pid
#DHCPDv6_PID=/var/run/dhcpd6.pid

# Additional options to start dhcpd with.
#       Don't use options -cf or -pf here; use DHCPD_CONF/ DHCPD_PID instead
#OPTIONS=""

# On what interfaces should the DHCP server (dhcpd) serve DHCP requests?
#       Separate multiple interfaces with spaces, e.g. "eth0 eth1".
INTERFACESv4="ens37"
INTERFACESv6=""
```

7. Setelah itu restart layanan DHCP jalankan:

```
sudo systemctl restart isc-dhcp-server
sudo systemctl status isc-dhcp-server
```

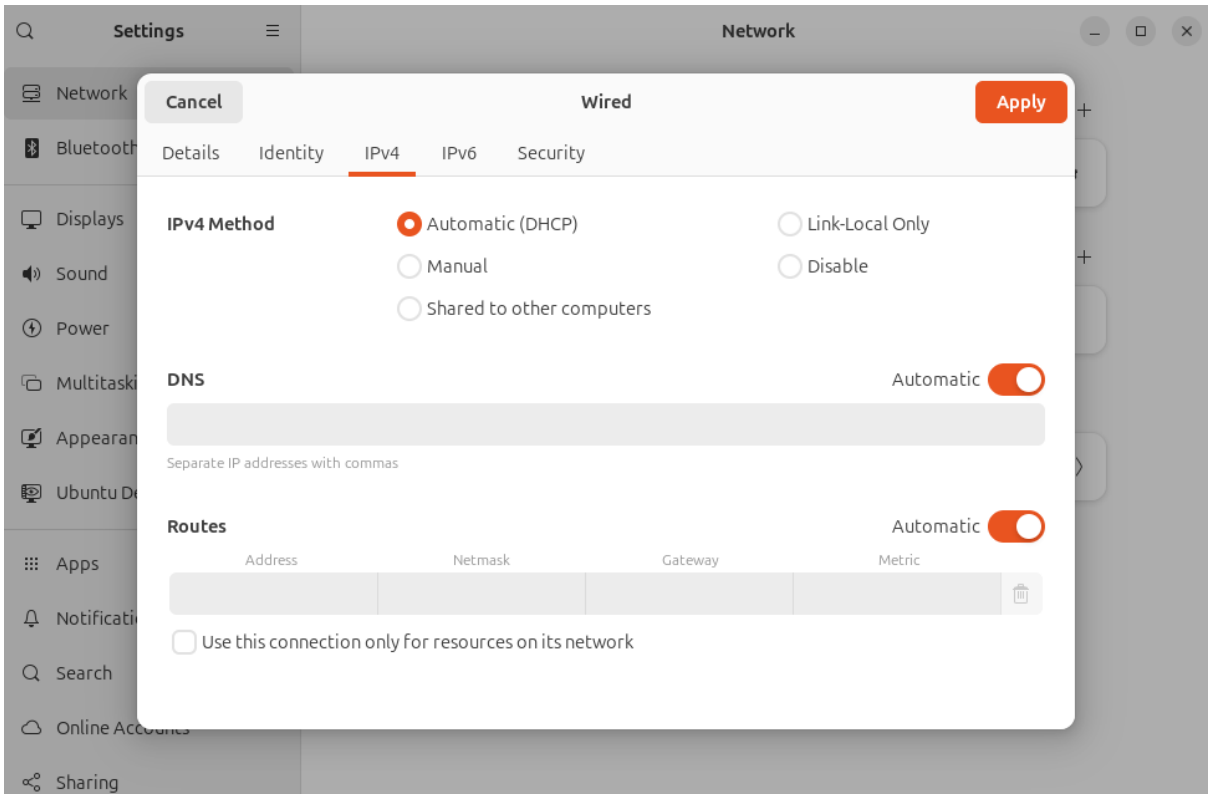
```
aisasilvana@aisasilvana:/etc/netplan$ sudo systemctl restart isc-dhcp-server.service
aisasilvana@aisasilvana:/etc/netplan$ sudo systemctl status isc-dhcp-server
● isc-dhcp-server.service - ISC DHCP IPv4 server
     Loaded: loaded (/usr/lib/systemd/system/isc-dhcp-server.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-04-21 04:46:36 UTC; 9s ago
       Docs: man:dhcpd(8)
   Main PID: 1667 (dhcpd)
      Tasks: 1 (limit: 4544)
     Memory: 3.8M (peak: 4.0M)
        CPU: 18ms
     CGroup: /system.slice/isc-dhcp-server.service
             └─1667 dhcpd -user dhcpd -group dhcpd -f -4 -pf /run/dhcp-server/dhcpd.pid -cf /etc/dhcp/dhcpd.conf ens37

Apr 21 04:46:36 aisasilvana dhcpd[1667]: PID file: /run/dhcp-server/dhcpd.pid
Apr 21 04:46:36 aisasilvana dhcpd[1667]: Wrote 2 leases to leases file.
Apr 21 04:46:36 aisasilvana sh[1667]: Wrote 2 leases to leases file.
Apr 21 04:46:36 aisasilvana dhcpd[1667]: Listening on LPF/ens37/00:0c:29:ff:f9:82/192.168.100.0/24
Apr 21 04:46:36 aisasilvana sh[1667]: Listening on LPF/ens37/00:0c:29:ff:f9:82/192.168.100.0/24
Apr 21 04:46:36 aisasilvana sh[1667]: Sending on   LPF/ens37/00:0c:29:ff:f9:82/192.168.100.0/24
Apr 21 04:46:36 aisasilvana sh[1667]: Sending on   Socket/fallback/fallback-net
Apr 21 04:46:36 aisasilvana dhcpd[1667]: Sending on   LPF/ens37/00:0c:29:ff:f9:82/192.168.100.0/24
Apr 21 04:46:36 aisasilvana dhcpd[1667]: Sending on   Socket/fallback/fallback-net
Apr 21 04:46:36 aisasilvana dhcpd[1667]: Server starting service.
aisasilvana@aisasilvana:/etc/netplan$ _
```

8. Sebelum melakukan konfigurasi DHCP, terlebih dahulu siapkan Virtual Machine (VM) pada VMware dengan spesifikasi sebagai berikut:

Device	Summary
Memory	4 GB
Processors	2
Hard Disk (SCSI)	15 GB
CD/DVD 2 (SATA)	Using file D:\ubuntu-24.04.3-...
CD/DVD (SATA)	Using file autoinst.iso
Floppy	Using file autoinst.flp
Network Adapter	Custom (VMnet1)
USB Controller	Present
Sound Card	Auto detect
Display	Auto detect

9. Setelah Virtual Machine selesai dibuat, buka menu Settings lalu Network pada Ubuntu desktop untuk melakukan konfigurasi jaringan. Pada bagian Wired → IPv4, pilih pengaturan: IPv4 Method : Automatic (DHCP)



10. Setelah konfigurasi jaringan diterapkan, buka terminal dan jalankan perintah berikut untuk melihat seluruh interface jaringan beserta alamat IP yang dimiliki.

```
aisasilvana@aisasilvana-VMware-Virtual-Platform:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:24:cc:b7 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.100.10/24 brd 192.168.100.255 scope global dynamic noprefixroute ens33
       valid_lft 596sec preferred_lft 596sec
    inet6 fe80::20c:29ff:fe24:ccb7/64 scope link
       valid_lft forever preferred_lft forever
aisasilvana@aisasilvana-VMware-Virtual-Platform:~$
```

Pada hasil pengecekan terlihat bahwa interface telah mendapatkan alamat IP secara otomatis yaitu inet 192.168.100.10/24 Hal ini menunjukkan bahwa koneksi jaringan berhasil dan Ubuntu Server sudah terhubung ke jaringan.

11. Selesai!